YZV202E Optimization for Data Science Energy Cost Optimization for European Countries

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***Abstract*—This paper presents optimization models for minimizing electricity generation costs across seven European countries over 2015–2023. Two models are developed: a Linear Program (LP) minimizing total generation cost including EU ETS carbon pricing, solved via PuLP/CBC Simplex and cross-validated against SciPy/HiGHS; and a convex Quadratic Program (QP) that additionally penalizes over-reliance on any single source, solved via CVXPY/OSQP and benchmarked against Newton interior-point and Conjugate Gradient methods. The LP yields cost-minimal corner solutions saturating cheapest sources, achieving 12–23 B$ in annual savings over historical German dispatch. The QP produces diversified mixes that better reflect real-world operational constraints. Carbon price sensitivity analysis shows that for Germany in 2022, the LP dispatch is invariant to carbon price as renewables saturate capacity limits regardless of pricing level, while the QP gradually phases out coal above $60–70/ton. A solver benchmark confirms SciPy/HiGHS is 1.8× faster than PuLP/CBC at identical solution quality.**

I. Introduction

Electricity generation planning directly affects both economic costs and carbon emissions. European countries operate diverse generation portfolios — coal, gas, nuclear, hydro, wind, and solar — each with distinct cost and emission characteristics. As the EU Emissions Trading System (ETS) carbon price rose from $8.45/ton in 2015 to $90.36/ton in 2023, the economics of fossil generation deteriorated significantly relative to renewables, raising the question of whether historical dispatch decisions have remained cost-optimal.

This paper applies Linear Programming (LP) and Quadratic Programming (QP) to determine the minimum-cost annual generation mix for seven European countries over 2015–2023. The LP minimizes total cost subject to demand balance and capacity constraints, solved via the Simplex method. The QP extends this with a quadratic penalty to prevent over-reliance on a single source, solved using CVXPY/OSQP with Newton and Conjugate Gradient. We compare both models against each other and perform sensitivity analysis on the carbon price.

II. Problem Formulation

A. Decision Variables

Let $x_i \geq 0$ denote the annual electricity generated from source i (GWh), where

$$S = \{\text{coal, gas, nuclear, hydro, wind, solar}\}.$$

B. Linear Program (LP)

$$\min_x \sum_{i \in S} 1000 \left(\text{LCOE}_{i,t} + C_{\text{CO}_2,t} \cdot E_i \right) x_i$$

Subject to:

$$\sum_{i \in S} x_i = D_{c,t}$$

$$0 \leq x_i \leq K_{i,c,t} \cdot \text{AF}_i \cdot 8760$$

where

- i denotes the electricity generation source,
- S is the set of generation sources,
- c denotes the country,
- t denotes the year,
- x_i (GWh) is the annual electricity generated from source i,
- $\text{LCOE}_{i,t}$ ($/MWh) is the year-specific levelized cost of electricity obtained from Lazard annual reports,
- $C_{\text{CO}_2,t}$ ($/ton) is the EU ETS carbon price,
- E_i (ton CO_2/MWh) is the lifecycle emission intensity of source i,
- $D_{c,t}$ (GWh) is the annual electricity demand,
- $K_{i,c,t}$ (GW) is the installed generation capacity,
- AF_i is the availability factor,
- 8760 is the number of hours in a year.

Since the objective function and constraints are linear, the LP attains its optimum at a vertex of the feasible polytope.

C. Quadratic Program (QP)

To prevent corner solutions and model diminishing returns, a quadratic penalty term is added:

$$\min_{x} \sum_{i \in S} \left[a_i x_i^2 + 1000 \left(\mathrm{LCOE}_{i,t} + C_{\mathrm{CO_2},t} \cdot E_i \right) x_i \right]$$

subject to the same constraints.

Here, a_i denotes the quadratic penalty coefficient associated with generation source i. The coefficients are sourced from PyPSA-Eur and Al-Roomi (2016). Since $a_i \geq 0$, the objective function is convex, ensuring a unique global optimum. The values used are

$$a_{\text{coal}} = 0.0015, \quad a_{\text{gas}} = 0.0020, \quad a_{\text{nuclear}} = 0.00001,$$

$$a_{\text{hydro}} = 0, \quad a_{\text{wind}} = 0.015, \quad a_{\text{solar}} = 0.010.$$

D. Availability Factors

Fixed availability factors from the U.S. DOE (2024) provide technology-based upper bounds independent of historical dispatch patterns:

TABLE I
AVAILABILITY FACTORS (U.S. DOE, 2024)

Source	AF
Nuclear	0.923
Gas	0.599
Coal	0.426
Hydro	0.345
Wind	0.343
Solar PV	0.234

III. DATASET & ALGORITHMIC WORKFLOW

A. Data Sources

Annual data for seven countries over 2015–2023 is drawn from three Eurostat datasets (Table II), supplemented by fixed parameters from external sources.

TABLE II
DATA SOURCES

Variable	Dataset	Unit
Demand D	NRG_CB_E	GWh
Capacity K_i	NRG_INF_EPC	MW
Generation P_i	NRG_BAL_PEH	GWh
$\mathrm{LCOE}_{i,t}$	Lazard Annual Reports	$/MWh
$C_{\mathrm{CO_2},t}$	ICAP ETS Database	$/ton
E_i	World Nuclear Association	t/MWh
AF_i	U.S. DOE (2024)	—

Coal/gas capacity decomposition. Eurostat reports coal and gas capacity jointly under SIEC code `CF`. We decompose using each year's generation shares:

$$K_{\text{coal}} = K_{\text{CF}} \cdot \frac{P_{\text{coal}}}{P_{\text{coal}} + P_{\text{gas}}}$$

Generation dataset. The full NRG_BAL_PEH dataset is $\sim$700 MB and was pre-filtered to relevant countries, years, SIEC codes, and unit (GWH).

B. Solver Pipeline

The LP is implemented in PuLP and solved by CBC Simplex, then cross-validated against SciPy `linprog` with the HiGHS backend. The QP is formulated in CVXPY's disciplined convex programming framework and solved by OSQP, an operator-splitting interior-point solver that factorizes the KKT system and iterates until primal-dual residuals fall below tolerance. Each country-year pair constitutes an independent optimization with six decision variables and seven constraints.

To analyze QP convergence behavior, a reference quadratic objective centered at the OSQP solution x^* is constructed and minimized using two SciPy methods: `trust-constr` (Newton interior-point, quadratic convergence) and `CG` (Conjugate Gradient with penalty relaxation, linear/superlinear convergence). Optimality gaps $f(x_k) - f^*$ are recorded at each iteration on a log scale.

IV. EXPERIMENTAL EVALUATION

A. LP Results — Germany 2022

Fig. 1 shows the LP optimal dispatch for Germany in 2022 at $84.42/ton carbon price. Wind and solar reach their availability-based capacity ceilings as the cheapest sources at 59 and 100 $/MWh respectively. Coal is significantly reduced due to its high unit cost of 187 $/MWh once carbon pricing is applied. Total optimal cost is 41,549 M$ with a system marginal cost of 85.2 $/MWh.

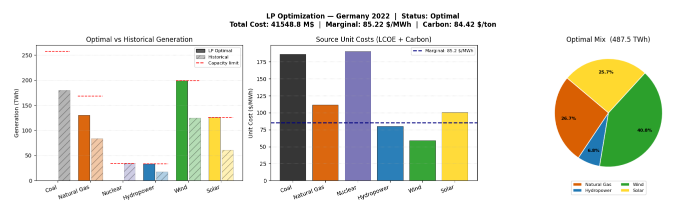

Fig. 1. LP optimization result for Germany 2022. Left: optimal vs historical generation with capacity limits. Center: unit costs with system marginal cost. Right: optimal generation mix.

B. Cost Trend — 2015–2023

Fig. 2 compares LP-optimal against historical cost for Germany over 2015–2023. Historical cost is computed using year-specific LCOE and actual carbon prices applied to observed generation quantities. The LP achieves savings of 12–23 B$ per year in every year of the study period. The gap widens after 2021 as the carbon price spike disproportionately penalizes the historically coal-heavy dispatch.

C. Cross-Country Comparison

Fig. 3 shows the LP-optimal mix and marginal cost for all seven countries in 2022. France's large nuclear fleet enables a low marginal cost, while Poland's coal-heavy capacity yields the highest marginal cost at 140.8 $/MWh. Germany achieves 85.2 $/MWh driven by its wind and solar fleet.

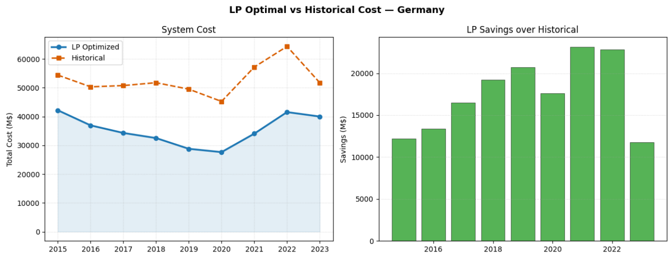

Fig. 2. LP-optimal vs historical total generation cost for Germany (2015–2023) with annual savings.

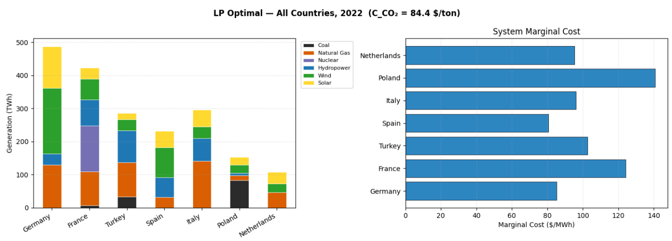

Fig. 3. LP-optimal generation mix (left) and system marginal cost (right) for all seven countries in 2022.

D. LP vs QP Comparison — Germany 2022

Fig. 4 compares the optimal dispatch produced by the LP and QP models for Germany in 2022. The two solutions are identical: wind, solar, hydro, and gas all hit their availability-based capacity ceilings under both formulations, with coal and nuclear at zero. The quadratic penalty term has negligible effect when capacity constraints are uniformly binding — the QP diversification benefit is may be more pronounced in countries with slack renewable capacity, where the LP would otherwise collapse to a single cheapest fossil source.

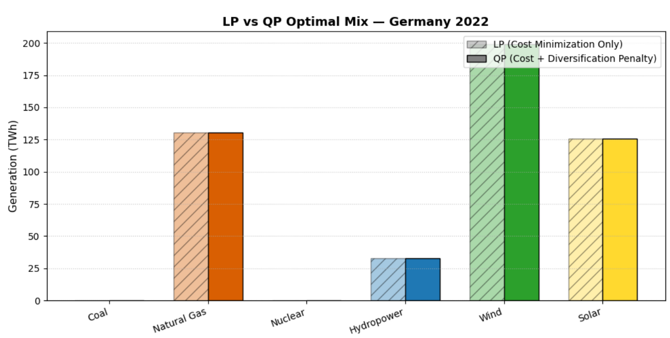

Fig. 4. LP vs QP optimal generation mix for Germany 2022.

E. Carbon Price Sensitivity

Figure 5 presents the effect of carbon price variations on the LP solution for Germany in 2022. The optimal generation mix remains unchanged across the entire carbon price range (0–200 \$/ton), indicating that renewable and other low-cost sources already operate at their capacity limits. As a result, higher carbon prices cannot alter the dispatch structure and only increase the cost of carbon-intensive generation.

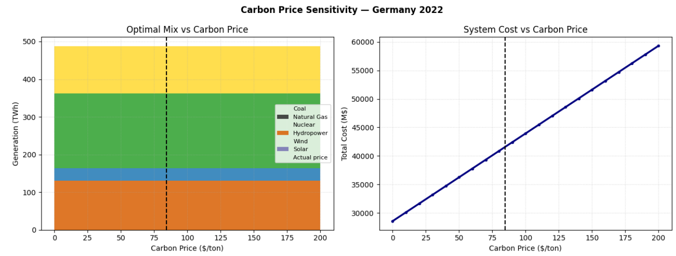

Fig. 5. Carbon price sensitivity analysis for Germany in 2022.

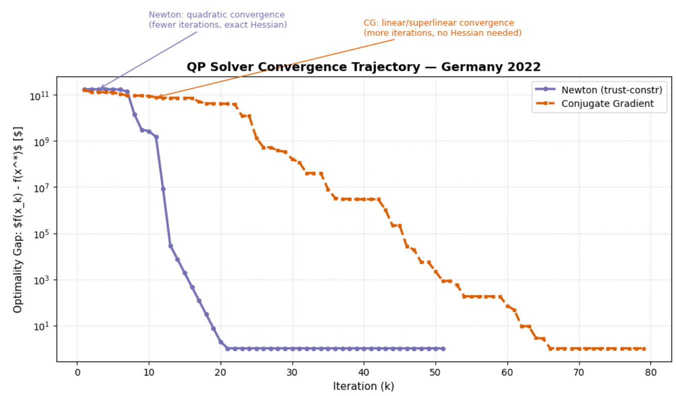

Fig. 6. Convergence comparison of Newton trust-region and Conjugate Gradient methods for the Germany 2022 QP instance.

Consequently, total system cost rises approximately linearly with carbon price, increasing from about 28.8 B\$ at zero carbon price to nearly 59 B\$ at 200 \$/ton. The dashed vertical line indicates the actual EU ETS carbon price level (approximately 85 \$/ton) observed in 2022.

F. Solver Benchmark

PuLP vs SciPy: Both solvers produce identical optimal costs across 63 country-year combinations (max difference <0.001 M\$). SciPy averages 3.51 ms versus 6.34 ms for PuLP — a 1.8× speedup.

G. Solver Convergence Analysis

Figure 6 compares Newton trust-region and Conjugate Gradient (CG) methods for the Germany 2022 QP instance. Newton reaches the optimum in approximately 20 iterations, exhibiting quadratic convergence. CG requires around 65 iterations but avoids explicit Hessian evaluations, resulting in lower per-iteration computational cost. Both methods converge to the same optimal solution.

V. Conclusion

This paper applied LP and QP to electricity dispatch optimization across seven European countries over 2015–2023. The LP confirms that cost-optimal dispatch would have significantly reduced coal and gas use, achieving 12–23 B\$ in annual savings over historical German generation costs. The QP produces more diversified mixes that better approximate real-world operational constraints, with coal phased out above \$60–70/ton carbon price.

Carbon price sensitivity reveals that for Germany in 2022, LP dispatch is invariant to carbon price — renewables already saturate their availability limits at any pricing level, indicating the energy transition has been completed on the supply side within the modelled capacity.

SciPy solves the LP 1.8× faster than PuLP at identical quality. Newton's method converges to QP optima in fewer iterations than CG, though CG's cheaper per-step cost makes it competitive in wall-clock time.

Limitations. Coal and gas capacity is decomposed from a joint Eurostat figure using generation shares, which remains an approximation. Availability factors from U.S. DOE may not fully reflect European grid conditions. LCOE values are global averages and do not capture country-specific financing costs.

GITHUB PAGE

The full implementation, including the Jupyter Notebook, data processing scripts, and pre-filtered datasets, is available at: https://github.com/asebyrm/Energy-Cost-Optimization

REFERENCES

[1] Eurostat, "Energy infrastructure and statistics (NRG_INF_EPC)," [Online]. Available: https://ec.europa.eu/eurostat/databrowser/view/nrg_inf_epc, accessed Apr. 2026.

[2] Eurostat, "Energy supply, transformation and consumption (NRG_CB_E)," [Online]. Available: https://ec.europa.eu/eurostat/databrowser/view/nrg_cb_e, accessed Apr. 2026.

[3] Eurostat, "Energy balances (NRG_BAL_PEH)," [Online]. Available: https://ec.europa.eu/eurostat/databrowser/view/nrg_bal_peh, accessed Apr. 2026.

[4] Lazard, "Levelized Cost of Energy Analysis (LCOE+), June 2025," [Online]. Available: https://www.lazard.com/media/5tlbhyla/lazards-lcoeplus-june-2025-vf.pdf, accessed Apr. 2026.

[5] International Carbon Action Partnership (ICAP), "Emissions Trading Scheme (ETS) price data," [Online]. Available: https://icapcarbonaction.com/en/ets-prices, accessed Apr. 2026.

[6] World Nuclear Association, "Carbon dioxide emissions from electricity generation," [Online]. Available: https://world-nuclear.org/information-library/energy-and-the-environment/carbon-dioxide-emissions-from-electricity, accessed Apr. 2026.

[7] PyPSA-Eur, "Renewable energy marginal costs," [Online]. Available: https://github.com/PyPSA/pypsa-eur/blob/master/data/custom_costs.csv, accessed Apr. 2026.

[8] A. R. Al-Roomi, "Economic Dispatch: 13-Units System Benchmark," [Online]. Available: http://www.al-roomi.org/economic-dispatch/13-units-system, 2016.

[9] S. Mitchell, M. O'Sullivan, and I. Dunning, "PuLP: A Linear Programming Toolkit for Python," The University of Auckland, Auckland, New Zealand, 2011.

[10] P. Virtanen et al., "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.

[11] U.S. Department of Energy, "What is Generation Capacity?" [Online]. Available: https://www.energy.gov/ne/articles/what-generation-capacity, accessed Apr. 2026.

[12] S. Diamond and S. Boyd, "CVXPY: A Python-Embedded Modeling Language for Convex Optimization," *J. Machine Learning Research*, vol. 17, no. 83, pp. 1–5, 2016.

[13] B. Stellato et al., "OSQP: An Operator Splitting Solver for Quadratic Programs," *Mathematical Programming Computation*, vol. 12, no. 4, pp. 637–672, 2020.